Progressive Education Society's
Modern College of Engineering, Pune-05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Department of S.Y. B. Tech [AIDS]

Assignment No: 05

SUBJECT: Java Programming

Name: Shravan Kalbhor
Class: SY BTech (AIDS)
Section: A
Roll Number: 26026

Title / Problem Statement :

Write a program to develop a basic employee management system that handles general employee data and includes specializations for different employee roles using inheritance. Demonstrate function overriding to customize behavior for specific roles. Make use ofthe 'this' keyword to refer to instance variables and super) to invoke the parent class constructor. Additionally, incorporate exception handling to manage invalid input or missing employee data gracefully during runtime (e.g., negative salary, empty name, or invalid role input).

Input :

```java
import java.util.Scanner;


// Parent Class
class Employee {
    String name;
    double salary;

    // Constructor
    Employee(String name, double salary) {
        this.name = name;
        this.salary = salary;
    }

    void display() {
        System.out.println("Employee Name: " + name);
        System.out.println("Salary: " + salary);
    }
}

// Child Class - Manager
class Manager extends Employee {

    Manager(String name, double salary) {
        super(name, salary); // calling parent constructor
    }

    @Override
    void display() {
        System.out.println("Role: Manager");
        super.display();
    }
}

// Child Class - Developer
class Developer extends Employee {

    Developer(String name, double salary) {
        super(name, salary);
    }
```

```java
    @Override
    void display() {
        System.out.println("Role: Developer");
        super.display();
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter name: ");
            String name = sc.nextLine();

            if (name.isEmpty()) {
                throw new Exception("Name cannot be empty!");
            }

            System.out.print("Enter salary: ");
            double salary = sc.nextDouble();

            if (salary < 0) {
                throw new Exception("Salary cannot be negative!");
            }

            sc.nextLine(); // clear buffer

            System.out.print("Enter role (Manager/Developer): ");
            String role = sc.nextLine();

            Employee emp;

            if (role.equalsIgnoreCase("Manager")) {
                emp = new Manager(name, salary);
            } else if (role.equalsIgnoreCase("Developer")) {
                emp = new Developer(name, salary);
            } else {
```

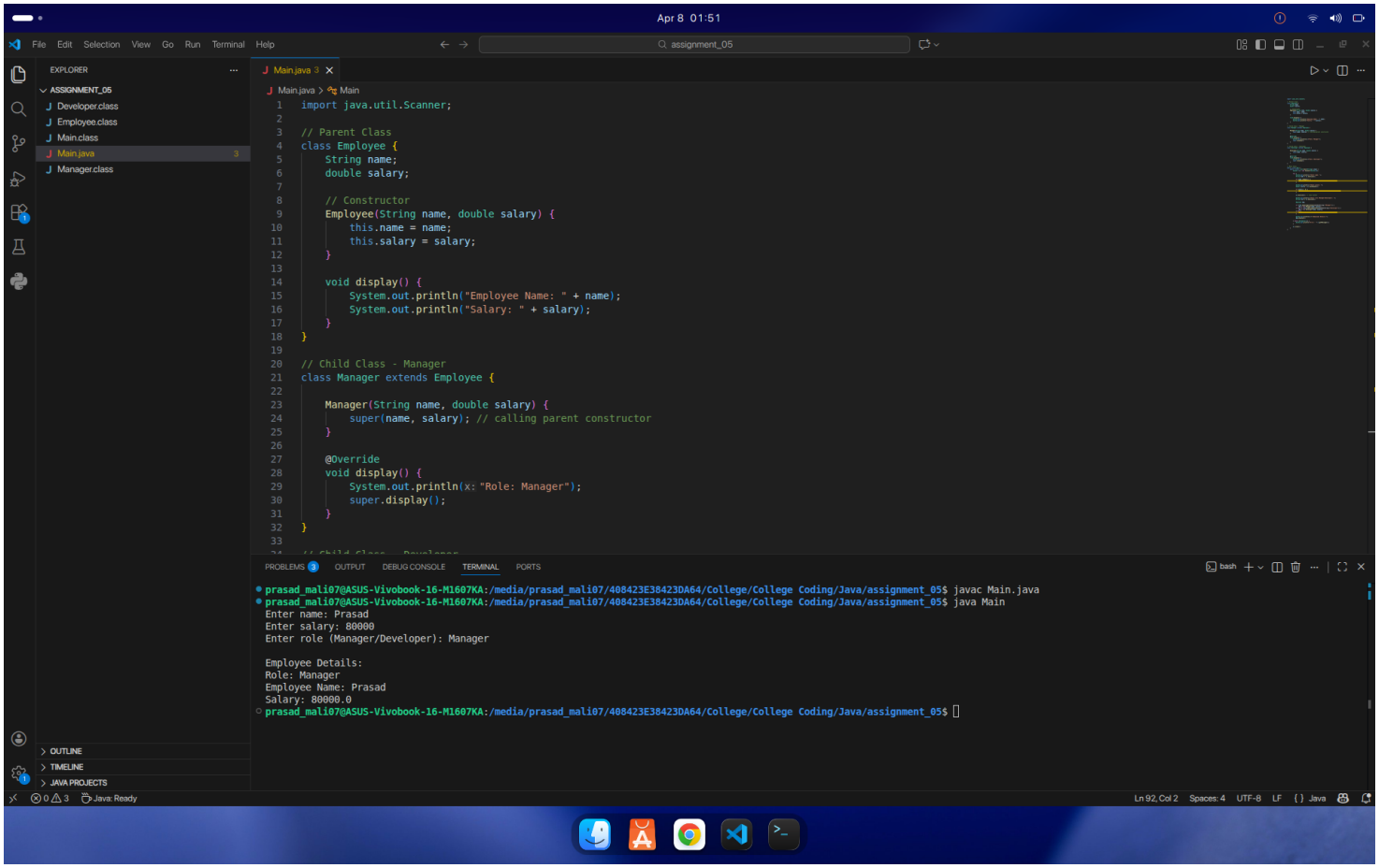
```
                throw new Exception("Invalid role entered!");
            }

            System.out.println("\nEmployee Details:");
            emp.display();

        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
        }

        sc.close();
    }
}
```

Output :


Apr 8 01:51
File Edit Selection View Go Run Terminal Help
assignment_05
EXPLORER
ASSIGNMENT_05
Developer.class
Employee.class
Main.class
Main.java 3
Manager.class
Main.java > Main
import java.util.Scanner;
// Parent Class
class Employee {
String name;
double salary;
// Constructor
Employee(String name, double salary) {
this.name = name;
this.salary = salary;
}
void display() {
System.out.println("Employee Name: " + name);
System.out.println("Salary: " + salary);
}
}
// Child Class - Manager
class Manager extends Employee {
Manager(String name, double salary) {
super(name, salary); // calling parent constructor
}
@Override
void display() {
System.out.println(x: "Role: Manager");
super.display();
}
}
PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_05$ javac Main.java
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_05$ java Main
Enter name: Prasad
Enter salary: 80000
Enter role (Manager/Developer): Manager
Employee Details:
Role: Manager
Employee Name: Prasad
Salary: 80000.0
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_05$
OUTLINE
TIMELINE
JAVA PROJECTS
Java: Ready
Ln 92, Col 2 Spaces: 4 UTF-8 LF { } Java